

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	G.Hima Bindu
Register Number	192511377

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Hima Bindu*

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.
4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

INTRODUCTION:

A media company wants to move its video streaming platform from traditional servers to the cloud. During live events, movie releases, or sports matches, millions of users may watch videos at the same time. If the platform is not designed properly, users may experience buffering, slow loading, or service failure. A cloud-based architecture provides scalability, flexibility, and high availability, making it suitable for large-scale video streaming.

1. Cloud Architecture:

A media company can migrate its video streaming platform to the cloud by using a scalable cloud architecture. The architecture includes cloud storage for storing video files, application servers for processing user requests, databases for storing user information, and a Content Delivery Network (CDN) for delivering videos from the nearest location. A Load Balancer is placed in front of the application servers to distribute incoming traffic, while Auto-Scaling automatically increases or decreases the number of servers based on user demand. This architecture can support millions of users with high availability and low latency.

2. Scalability and Caching Techniques:

Scalability allows the cloud platform to increase resources whenever the number of users grows. During live events or popular movie releases, additional servers are automatically added to handle the increased traffic. Caching improves performance by storing frequently accessed videos and user session data in memory. The CDN stores copies of popular videos at multiple edge locations so users receive content from the nearest server. This reduces buffering, decreases response time, and improves the overall streaming experience.

3. Load Balancing and Auto-Scaling:

Load balancing distributes incoming user requests equally among multiple application servers, preventing any single server from becoming overloaded. It continuously checks server health and redirects traffic if a server fails. Auto-scaling monitors CPU usage, memory utilization, and network traffic. When user traffic increases, new servers are launched automatically, and when traffic decreases, unnecessary servers are removed. Together, load balancing and auto-scaling ensure smooth video streaming, better performance, and uninterrupted service during peak workloads.

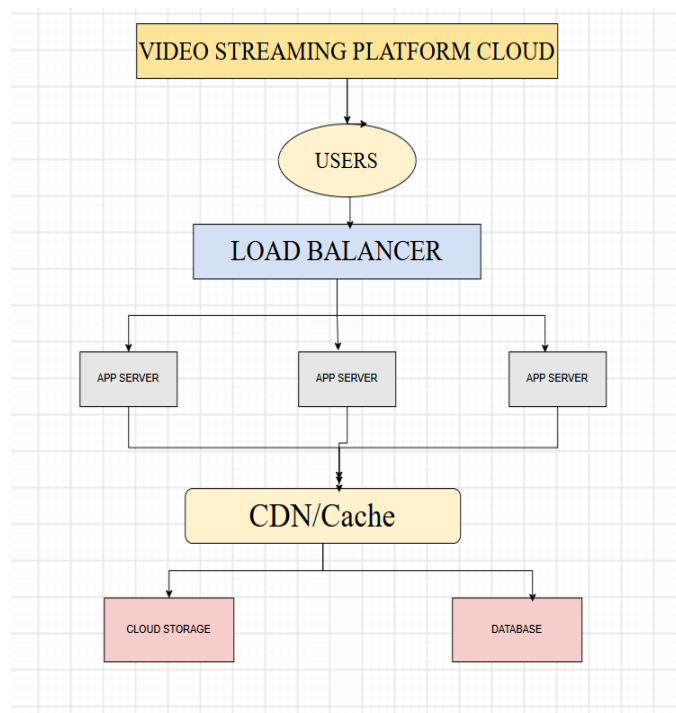
4. Cost Optimization Strategies:

Cloud costs can be reduced by using auto-scaling, which ensures that servers run only when needed. Frequently watched videos should be stored in high-speed storage, while rarely accessed content can be

moved to low-cost archival storage. A CDN reduces bandwidth usage by serving content from nearby locations instead of the main server. Video compression techniques help reduce storage requirements and network usage. Regular monitoring of cloud resources and removing unused instances also help minimize operational costs.

5. High Availability and Reliability:

To maintain continuous service, the streaming platform should deploy application servers across multiple availability zones or cloud regions. Data should be backed up regularly, and databases should use replication to prevent data loss. Health monitoring tools should continuously check server performance and automatically replace failed instances. This ensures that even if one server or region fails, users can continue watching videos without interruption, providing a reliable and fault-tolerant streaming platform.



2.A fintech startup is deploying applications using containers across multiple cloud providers.

They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

INTRODUCTION:

A fintech startup is deploying its banking and payment applications using containers across multiple cloud providers such as AWS, Microsoft Azure, and Google Cloud. Since the applications run on different cloud platforms, the company faces challenges such as maintaining consistency, managing deployments, ensuring

service communication, and handling increasing user traffic. A multi-cloud strategy with container orchestration provides a reliable and scalable solution.

1. Container Orchestration and Multi-Cloud Strategy

A fintech startup can use Docker containers to package its applications and Kubernetes as the container orchestration platform. Kubernetes manages container deployment, scheduling, scaling, and monitoring across multiple cloud providers such as AWS, Microsoft Azure, and Google Cloud. A central container registry stores application images so that the same version of the application is deployed in every cloud environment. This ensures consistency, simplifies deployment, and reduces configuration differences between cloud providers.

2. Maintaining Consistency and Service Discovery

To maintain consistency, the startup should use the same container images, configuration files, and deployment policies across all cloud platforms. Infrastructure as Code (IaC) tools like Terraform can create identical environments in different clouds. Kubernetes provides service discovery by assigning a unique DNS name to every service, allowing containers to communicate without using fixed IP addresses. This enables reliable communication between microservices even when containers are restarted or moved to another server.

3. Scaling and Deployment Management

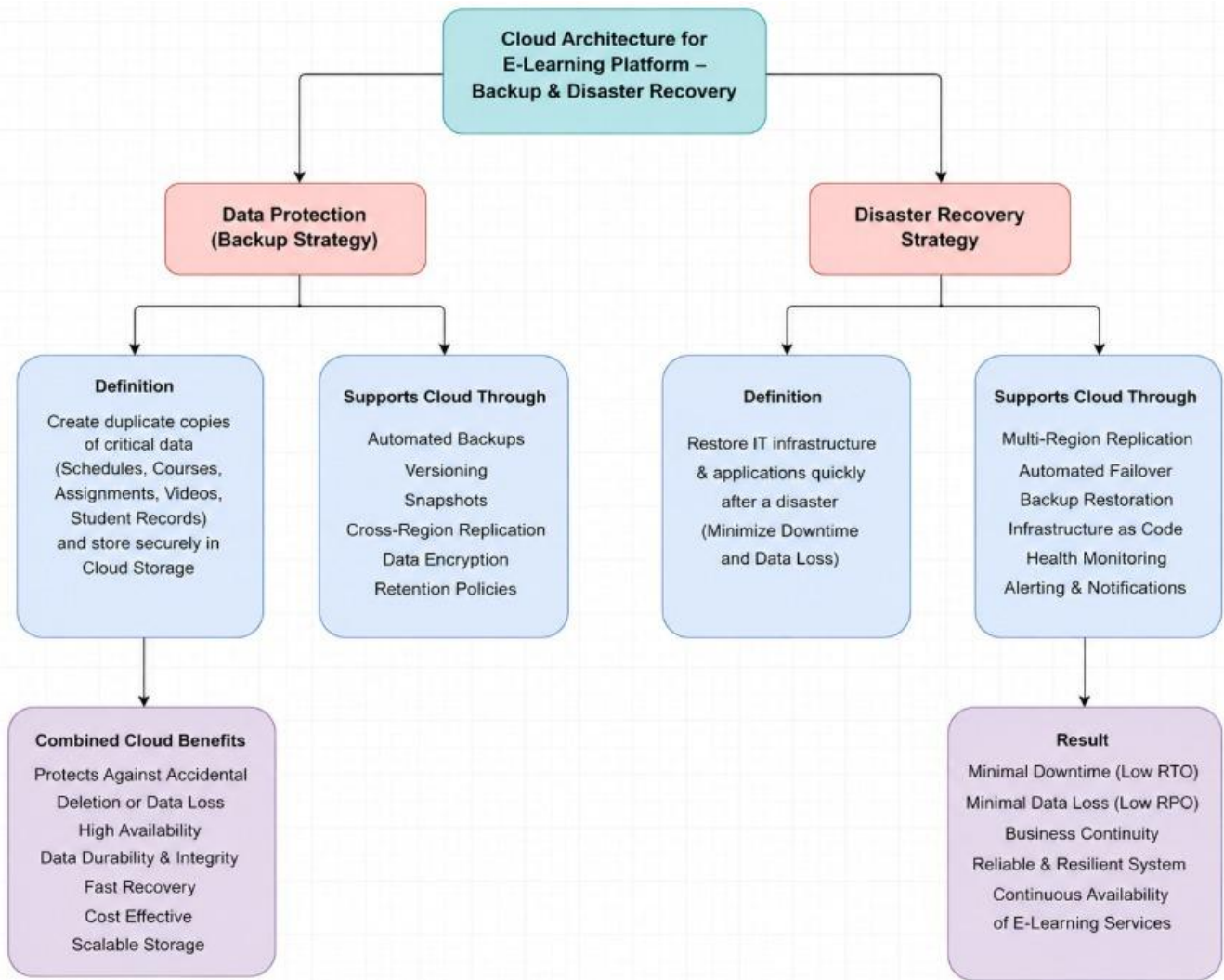
Kubernetes supports automatic scaling using the Horizontal Pod Autoscaler (HPA). It continuously monitors CPU usage, memory usage, and network traffic, then increases or decreases the number of container instances based on demand. Load balancing distributes incoming requests equally among all available containers, ensuring better performance and preventing server overload. Kubernetes also performs rolling updates and automatically replaces failed containers, allowing application updates without downtime.

4. Benefits of Multi-Cloud Deployment

A multi-cloud deployment provides several advantages for fintech applications. It improves high availability because if one cloud provider experiences a failure, applications continue running on another cloud. It prevents vendor lock-in, allowing the company to choose the best services from different providers. Multi-cloud also improves disaster recovery, scalability, performance, and compliance by deploying applications closer to users and storing data according to regional regulations.

5. Risks of Multi-Cloud Deployment

Although multi-cloud offers many benefits, it also introduces several challenges. Managing applications across multiple cloud providers increases operational complexity and requires skilled administrators. Security policies must be implemented consistently across all environments to avoid vulnerabilities. Data synchronization between clouds can be difficult, and communication between different cloud providers may increase network latency. In addition, monitoring, maintenance, and infrastructure costs may increase because multiple cloud platforms need to be managed simultaneously.



3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.

Provide steps to ensure business continuity during failures.

INTRODUCTION:

An e-learning platform stores student records, course materials, videos, assignments, and examination data in the cloud. If important files are accidentally deleted or a cloud failure occurs, students and teachers may lose access to learning resources. To avoid such problems, the organization should implement a strong Backup and Disaster Recovery (DR) strategy using cloud-native backup services, data replication, and object versioning. This ensures that data can be recovered quickly with minimum downtime.

1. Cloud-Native Backup and Disaster Recovery Solution

The e-learning platform should use a cloud-native backup and disaster recovery solution to protect important data such as student records, course materials, videos, assignments, and examination data. All files should be stored in cloud object storage, while automatic backups should be scheduled daily or every few hours. Data replication should be enabled to create copies of data in different cloud regions or availability zones, ensuring that if one region fails, another region continues providing services. Object versioning should also be enabled to maintain multiple versions of files so that accidentally deleted or modified files can be restored quickly. In addition, a disaster recovery site should be maintained in another geographical region with replicated databases, backup copies, application servers, and storage to ensure uninterrupted access during failures.

2. Backup Strategy, Replication, and Versioning

The organization should follow the 3-2-1 Backup Rule, which means keeping three copies of important data, storing them in two different locations, and maintaining one copy in another cloud region. Full database backups should be taken daily, incremental backups should be performed every hour, and disaster recovery testing should be carried out regularly. Data replication should continuously copy information from the primary cloud region to a secondary region. Synchronous replication is suitable for critical databases because it updates both locations simultaneously, while asynchronous replication is suitable for videos and learning materials where a slight delay is acceptable. Versioning keeps previous copies of files, allowing administrators to recover earlier versions if files are accidentally deleted, corrupted, or modified, thereby protecting against accidental deletion and ransomware attacks.

3. Recovery Time Objective (RTO) and Recovery Point Objective (RPO)

Recovery Time Objective (RTO) is the maximum acceptable time required to restore services after a failure. It can be achieved by maintaining standby servers, using automated backup restoration, deploying disaster recovery servers, configuring automatic failover, and continuously monitoring system health. For example, if the RTO is set to 30 minutes, the platform should become operational within that time after a failure. Recovery Point Objective (RPO) represents the maximum amount of data that can be lost during a disaster. It is achieved through frequent automated backups, continuous database replication, object versioning, and real-time synchronization between cloud regions. If backups occur every 10 minutes, only the last ten minutes of data may be lost during a failure.

4. Business Continuity During Failures

To ensure business continuity, the organization should perform continuous automated backups of all educational data and enable replication of application servers and databases to another cloud region. Multiple versions of important files should be maintained using versioning so that deleted or modified files can be restored easily. Automatic failover should redirect user traffic to the disaster recovery site whenever the primary cloud becomes unavailable. Regular disaster recovery testing should be conducted to verify that backup restoration procedures work correctly. Continuous monitoring of CPU usage, storage, network traffic, database health, and backup status helps identify problems before they become critical. Backup data should also be protected using encryption, Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC), and regular security updates to prevent unauthorized access and ensure secure recovery during disasters.

5. Benefits and Challenges

This backup and disaster recovery solution prevents permanent data loss, enables quick recovery of deleted files, improves system reliability, reduces downtime, and ensures continuous access for students and teachers. Cloud-native backup, replication, and versioning also improve data security and availability. However, implementing this solution has some challenges, including increased storage costs, higher operational expenses for maintaining disaster recovery infrastructure, additional network bandwidth consumption for replication, and the need for regular backup verification and disaster recovery testing to ensure the system remains reliable.

4.A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

1. Hybrid Cloud Architecture and Secure Communication

A government agency can adopt a hybrid cloud architecture by keeping sensitive applications and confidential data, such as citizen records, financial information, and national security data, in the on-premises data center. Non-sensitive applications like public websites, e-governance portals, and online citizen services can be deployed in the public cloud. Both environments are connected using a secure VPN (Virtual Private Network) or a dedicated private network to ensure safe communication. Firewalls, network segmentation, intrusion detection systems (IDS), and intrusion prevention systems (IPS) should be implemented to monitor and protect network traffic. All data transferred between the cloud and on-premises environment should be encrypted using TLS (Transport Layer Security) or IPsec protocols to prevent unauthorized access and ensure secure communication.

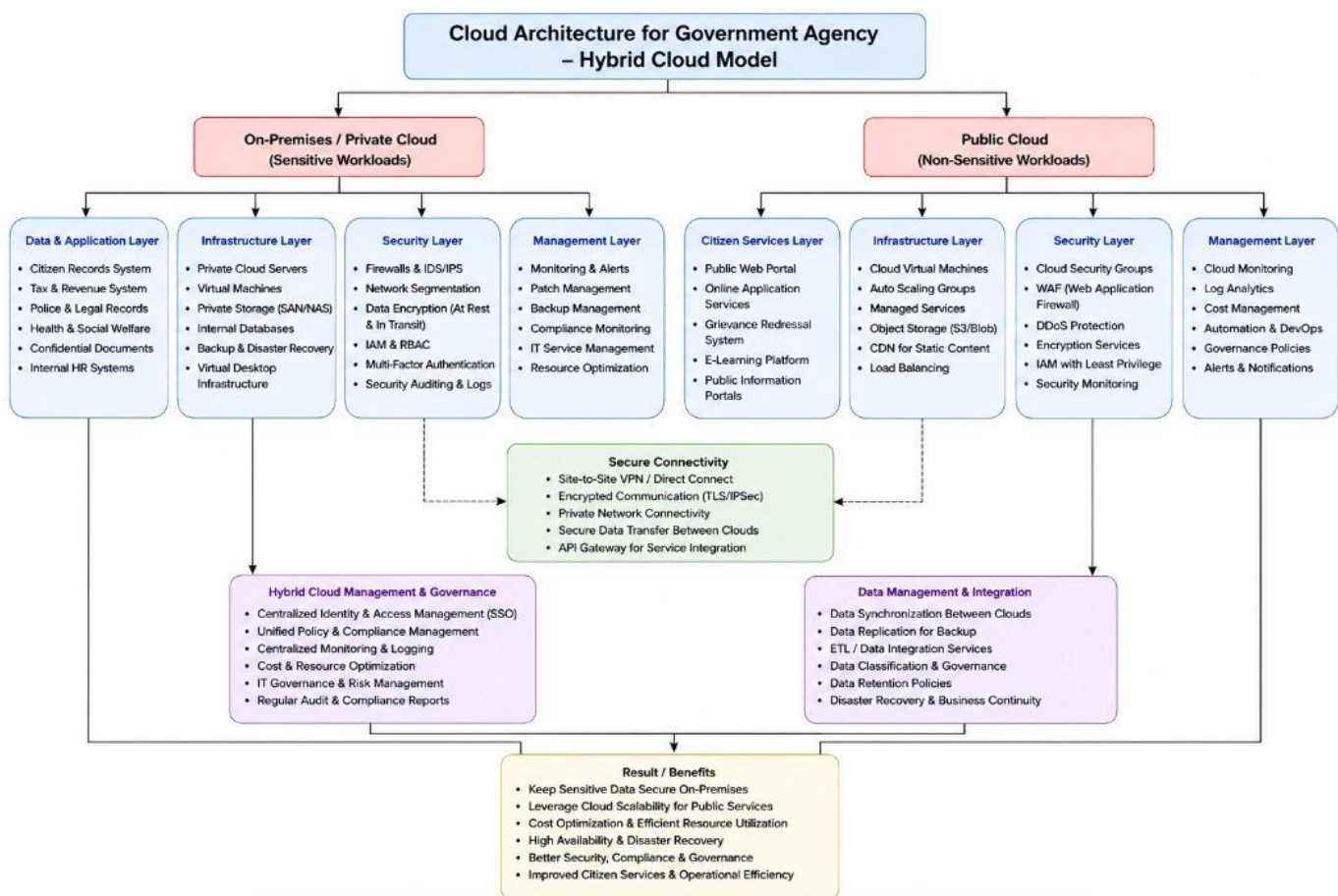
2. Identity Management and Encryption

Identity management is essential to control user access in a hybrid cloud environment. The organization should implement a centralized Identity and Access Management (IAM) system so that users can securely access both cloud and on-premises resources using Single Sign-On (SSO). Multi-Factor Authentication (MFA) should be enabled to provide additional security by requiring users to verify their identity through passwords, OTPs, or biometric authentication. Role-Based Access Control (RBAC) ensures that users can access only the resources required for their job roles. To protect sensitive government data, encryption should be applied to both data at rest and data in transit. Data stored in databases and cloud storage should be encrypted using AES-256, while data moving between environments

should be protected using TLS or VPN encryption. Encryption keys should be securely managed through a Key Management Service (KMS) with regular key rotation.

3.Challenges in Managing Hybrid Cloud Environments

Managing a hybrid cloud environment is more complex than managing a single cloud because resources are distributed across both on-premises infrastructure and the public cloud. Maintaining consistent security policies, ensuring compliance with government regulations, and synchronizing data between both environments require careful planning. Network latency may occur when applications communicate across different environments, and monitoring multiple platforms can increase administrative effort. In addition, maintaining separate infrastructure, cloud services, backup systems, and skilled personnel can increase operational costs. However, despite these challenges, a hybrid cloud provides better scalability, higher availability, improved disaster recovery, stronger security, and greater flexibility while ensuring that sensitive government data remains protected and compliant with regulatory requirements.



5.A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

1. Multi-Region Deployment Strategy

A SaaS provider should deploy its application across multiple cloud regions to ensure continuous availability. Each region should have its own application servers, databases, storage, and load balancers. A Global Load Balancer or DNS-based traffic manager directs users to the nearest available region. If one region becomes unavailable due to a hardware failure or natural disaster, traffic is automatically redirected to another healthy region without interrupting the service.

2. Data Replication and Failover Mechanisms

The application database should use data replication to maintain copies of data in multiple regions. Critical data can use synchronous replication for real-time updates, while less critical data can use asynchronous replication. Automatic failover should be configured using health checks and standby servers. When the primary region fails, the secondary region automatically becomes active, ensuring users can continue accessing the application with minimal downtime and data loss.

3. Monitoring and Alerting Strategies

Cloud monitoring tools should continuously monitor system performance, including CPU utilization, memory usage, storage, network traffic, application response time, and database health. Automated alerts should be configured to notify administrators through email, SMS, or messaging applications whenever resource usage exceeds the defined threshold or any service becomes unavailable. Centralized log management should also be used to detect application errors, security threats, and unusual activities quickly.

4. High Availability and Fault Tolerance

High availability is achieved by deploying identical application components in different cloud regions so that if one region fails, another region immediately continues providing services. Fault tolerance is achieved through redundant servers, replicated databases, backup storage, and automatic failover mechanisms. Regular backup verification, disaster recovery testing, and health checks should be performed to ensure that the application can recover quickly from failures and continue operating without affecting users.

